Bayesian Conjugate Priors — One■Page Cheat Sheet

1. Core pattern:

Posterior ∝ Likelihood × Prior.

2. Conjugate families:

Bernoulli/Binomial → Beta

Poisson → Gamma

Normal mean → Normal

Multinomial → Dirichlet

3. Generic update rules:

$\text{Beta}(\alpha,\beta) + \text{Binomial}(S,n-S) \to \text{Beta}(\alpha+S,\beta+n-S)$

$\text{Gamma}(\alpha,\beta) + \text{Poisson}(S,n) \to \text{Gamma}(\alpha+S,\beta+n)$

4. Posterior mean and MAP:

$\text{Mean} = \alpha'/\beta'$

$\text{MAP} = (\alpha'-1)/(\beta')$ when $\alpha'>1$

5. Template:

Prior: $\pi(\theta)$

Likelihood: $L(\theta)$

Posterior: $\pi(\theta|x) \propto \pi(\theta)L(\theta)$

This is a shortened PDF version of the cheat sheet.